Practical 10

Aim:- Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:-

```
 GNU nano 7.2
include <stdio.h>
include <stdlib.h>

nt main() {
   int req[] = {98,183,41,122,14,124,65,67};
   int n = 8;
   int head = 53;
   int visited[8] = {0};
   int total_movement = 0;

   int i, j, min_dist, index;

   for(i=0;i<n;i++) {
       min_dist = 1000;

       for(j=0;j<n;j++) {
           if(!visited[j]) {
               int dist = abs(head - req[j]);

               if(dist < min_dist) {
                   min_dist = dist;
                   index = j;
               }
           }
       }

       total_movement += min_dist;
       head = req[index];
       visited[index] = 1;
   }

   printf("\nTotal Head Movement = %d\n", total_movement)

   return 0;
```

Output:-

```
kalash@KALASH: ~

kalash@KALASH:~$ nano sstf.c
kalash@KALASH:~$ gcc sstf.c -o sstf
kalash@KALASH:~$ ./sstf

Total Head Movement = 323
```